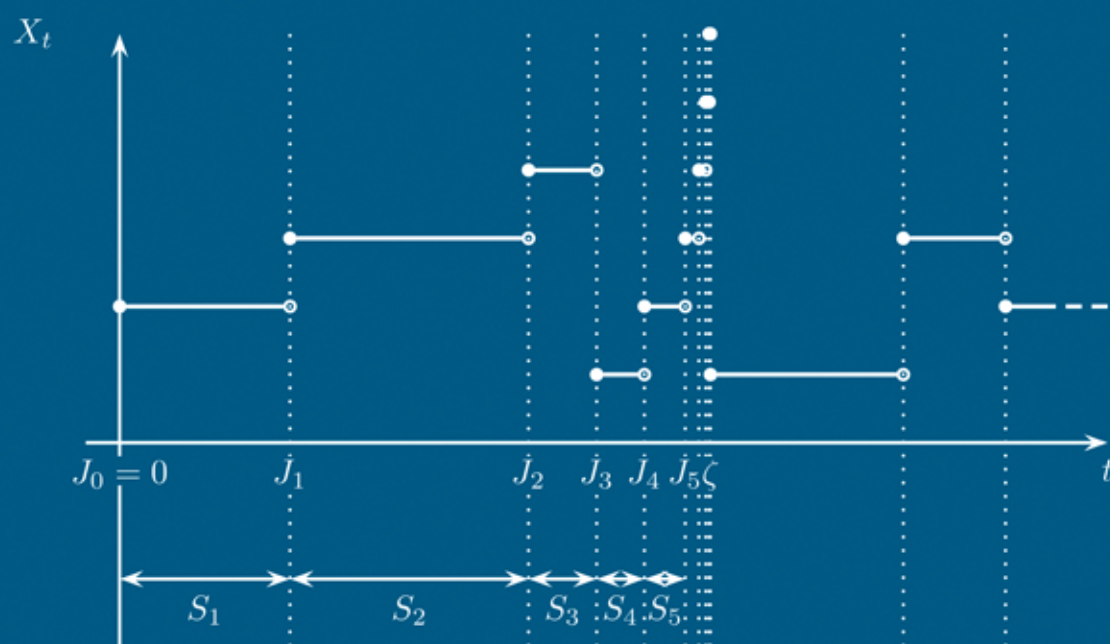


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Markov Chains

J.R. Norris

Markov Chains

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Markov Chains

J. R. Norris
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For my parents

Contents

Preface	ix
Introduction	xiii
1. Discrete-time Markov chains	1
1.1 Definition and basic properties	1
1.2 Class structure	10
1.3 Hitting times and absorption probabilities	12
1.4 Strong Markov property	19
1.5 Recurrence and transience	24
1.6 Recurrence and transience of random walks	29
1.7 Invariant distributions	33
1.8 Convergence to equilibrium	40
1.9 Time reversal	47
1.10 Ergodic theorem	52
1.11 Appendix: recurrence relations	57
1.12 Appendix: asymptotics for $n!$	58
2. Continuous-time Markov chains I	60
2.1 Q -matrices and their exponentials	60
2.2 Continuous-time random processes	67
2.3 Some properties of the exponential distribution	70

Contents

2.4 Poisson processes	73
2.5 Birth processes	81
2.6 Jump chain and holding times	87
2.7 Explosion	90
2.8 Forward and backward equations	93
2.9 Non-minimal chains	103
2.10 Appendix: matrix exponentials	105
3. Continuous-time Markov chains II	108
3.1 Basic properties	108
3.2 Class structure	111
3.3 Hitting times and absorption probabilities	112
3.4 Recurrence and transience	114
3.5 Invariant distributions	117
3.6 Convergence to equilibrium	121
3.7 Time reversal	123
3.8 Ergodic theorem	125
4. Further theory	128
4.1 Martingales	128
4.2 Potential theory	134
4.3 Electrical networks	151
4.4 Brownian motion	159
5. Applications	170
5.1 Markov chains in biology	170
5.2 Queues and queueing networks	179
5.3 Markov chains in resource management	192
5.4 Markov decision processes	197
5.5 Markov chain Monte Carlo	206
6. Appendix: probability and measure	217
6.1 Countable sets and countable sums	217
6.2 Basic facts of measure theory	220
6.3 Probability spaces and expectation	222
6.4 Monotone convergence and Fubini's theorem	223
6.5 Stopping times and the strong Markov property	224
6.6 Uniqueness of probabilities and independence of σ -algebras	228
Further reading	232
Index	234

Preface

Markov chains are the simplest mathematical models for random phenomena evolving in time. Their simple structure makes it possible to say a great deal about their behaviour. At the same time, the class of Markov chains is rich enough to serve in many applications. This makes Markov chains the first and most important examples of random processes. Indeed, the whole of the mathematical study of random processes can be regarded as a generalization in one way or another of the theory of Markov chains.

This book is an account of the elementary theory of Markov chains, with applications. It was conceived as a text for advanced undergraduates or master's level students, and is developed from a course taught to undergraduates for several years. There are no strict prerequisites but it is envisaged that the reader will have taken a course in elementary probability. In particular, measure theory is not a prerequisite.

The first half of the book is based on lecture notes for the undergraduate course. Illustrative examples introduce many of the key ideas. Careful proofs are given throughout. There is a selection of exercises, which forms the basis of classwork done by the students, and which has been tested over several years. [Chapter 1](#) deals with the theory of discrete-time Markov chains, and is the basis of all that follows. You must begin here. The material is quite straightforward and the ideas introduced permeate the whole book. The basic pattern of [Chapter 1](#) is repeated in [Chapter 3](#) for continuous-time chains, making it easy to follow the development by analogy. In between, [Chapter 2](#) explains how to set up the theory of continuous-

Preface

time chains, beginning with simple examples such as the Poisson process and chains with finite state space.

The second half of the book comprises three independent chapters intended to complement the first half. In some sections the style is a little more demanding. [Chapter 4](#) introduces, in the context of elementary Markov chains, some of the ideas crucial to the advanced study of Markov processes, such as martingales, potentials, electrical networks and Brownian motion. [Chapter 5](#) is devoted to applications, for example to population growth, mathematical genetics, queues and networks of queues, Markov decision processes and Monte Carlo simulation. [Chapter 6](#) is an appendix to the main text, where we explain some of the basic notions of probability and measure used in the rest of the book and give careful proofs of the few points where measure theory is really needed.

The following paragraph is directed primarily at an instructor and assumes some familiarity with the subject. Overall, the book is more focused on the Markovian context than most other books dealing with the elementary theory of stochastic processes. I believe that this restriction in scope is desirable for the greater coherence and depth it allows. The treatment of discrete-time chains in [Chapter 1](#) includes the calculation of transition probabilities, hitting probabilities, expected hitting times and invariant distributions. Also treated are recurrence and transience, convergence to equilibrium, reversibility, and the ergodic theorem for long-run averages. All the results are proved, exploiting to the full the probabilistic viewpoint. For example, we use excursions and the strong Markov property to obtain conditions for recurrence and transience, and convergence to equilibrium is proved by the coupling method. In [Chapters 2](#) and [3](#) we proceed via the jump chain/holding time construction to treat all right-continuous, minimal continuous-time chains, and establish analogues of all the main results obtained for discrete time. No conditions of uniformly bounded rates are needed. The student has the option to take [Chapter 3](#) first, to study the *properties* of continuous-time chains before the technically more demanding *construction*. We have left measure theory in the background, but the proofs are intended to be rigorous, or very easily made rigorous, when considered in measure-theoretic terms. Some further details are given in [Chapter 6](#).

It is a pleasure to acknowledge the work of colleagues from which I have benefitted in preparing this book. The course on which it is based has evolved over many years and under many hands – I inherited parts of it from Martin Barlow and Chris Rogers. In recent years it has been given by Doug Kennedy and Colin Sparrow. Richard Gibbens, Geoffrey Grim-

Preface

mett, Frank Kelly and Gareth Roberts gave expert advice at various stages. Meena Lakshmanan, Violet Lo and David Rose pointed out many typos and ambiguities. Brian Ripley and David Williams made constructive suggestions for improvement of an early version.

I am especially grateful to David Tranah at Cambridge University Press for his suggestion to write the book and for his continuing support, and to Sarah Shea-Simonds who typeset the whole book with efficiency, precision and good humour.

Cambridge, 1996

James Norris

Introduction

This book is about a certain sort of random process. The characteristic property of this sort of process is that it retains *no memory* of where it has been in the past. This means that only the current state of the process can influence where it goes next. Such a process is called a *Markov process*. We shall be concerned exclusively with the case where the process can assume only a finite or countable set of states, when it is usual to refer to it as a *Markov chain*.

Examples of Markov chains abound, as you will see throughout the book. What makes them important is that not only do Markov chains model many phenomena of interest, but also the lack of memory property makes it possible to predict how a Markov chain may behave, and to compute probabilities and expected values which quantify that behaviour. In this book we shall present general techniques for the analysis of Markov chains, together with many examples and applications. In this introduction we shall discuss a few very simple examples and preview some of the questions which the general theory will answer.

We shall consider chains both in *discrete time*

$$n \in \mathbb{Z}^+ = \{0, 1, 2, \dots\}$$

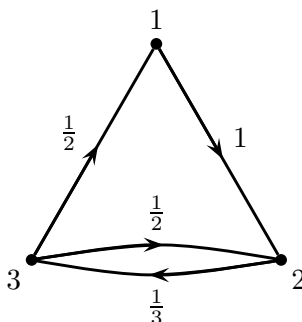
and *continuous time*

$$t \in \mathbb{R}^+ = [0, \infty).$$

The letters n, m, k will always denote integers, whereas t and s will refer to real numbers. Thus we write $(X_n)_{n \geq 0}$ for a discrete-time process and $(X_t)_{t \geq 0}$ for a continuous-time process.

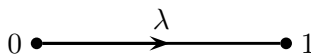
Markov chains are often best described by diagrams, of which we now give some simple examples:

(i) (*Discrete time*)



You move from state 1 to state 2 with probability 1. From state 3 you move either to 1 or to 2 with equal probability $1/2$, and from 2 you jump to 3 with probability $1/3$, otherwise stay at 2. We might have drawn a loop from 2 to itself with label $2/3$. But since the total probability on jumping from 2 must equal 1, this does not convey any more information and we prefer to leave the loops out.

(ii) (*Continuous time*)

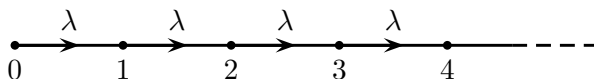


When in state 0 you wait for a random time with exponential distribution of parameter $\lambda \in (0, \infty)$, then jump to 1. Thus the density function of the waiting time T is given by

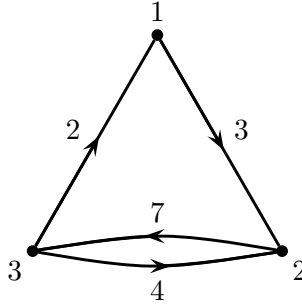
$$f_T(t) = \lambda e^{-\lambda t} \quad \text{for } t \geq 0.$$

We write $T \sim E(\lambda)$ for short.

(iii) (*Continuous time*)



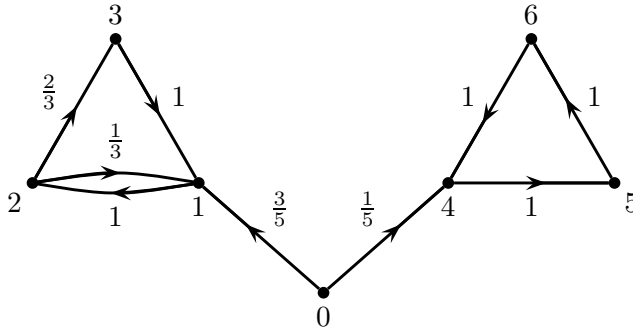
Here, when you get to 1 you do not stop but after another independent exponential time of parameter λ jump to 2, and so on. The resulting process is called the *Poisson process of rate λ* .



(iv) (*Continuous time*)

In state 3 you take two independent exponential times $T_1 \sim E(2)$ and $T_2 \sim E(4)$; if T_1 is the smaller you go to 1 after time T_1 , and if T_2 is the smaller you go to 2 after time T_2 . The rules for states 1 and 2 are as given in examples (ii) and (iii). It is a simple matter to show that the time spent in 3 is exponential of parameter $2 + 4 = 6$, and that the probability of jumping from 3 to 1 is $2/(2 + 4) = 1/3$. The details are given later.

(v) (*Discrete time*)



We use this example to anticipate some of the ideas discussed in detail in [Chapter 1](#). The states may be partitioned into *communicating classes*, namely $\{0\}$, $\{1, 2, 3\}$ and $\{4, 5, 6\}$. Two of these classes are *closed*, meaning that you cannot escape. The closed classes here are *recurrent*, meaning that you return again and again to every state. The class $\{0\}$ is *transient*. The class $\{4, 5, 6\}$ is *periodic*, but $\{1, 2, 3\}$ is not. We shall show how to establish the following facts by solving some simple linear equations. You might like to try from first principles.

- (a) Starting from 0, the probability of hitting 6 is $1/4$.
- (b) Starting from 1, the probability of hitting 3 is 1.
- (c) Starting from 1, it takes on average three steps to hit 3.
- (d) Starting from 1, the long-run proportion of time spent in 2 is $3/8$.

Let us write $p_{ij}^{(n)}$ for the probability starting from i of being in state j after n steps. Then we have:

(e) $\lim_{n \rightarrow \infty} p_{01}^{(n)} = 9/32;$

(f) $p_{04}^{(n)}$ does not converge as $n \rightarrow \infty;$

(g) $\lim_{(n) \rightarrow \infty} p_{04}^{(3n)} = 1/124.$

1

Discrete-time Markov chains

This chapter is the foundation for all that follows. Discrete-time Markov chains are defined and their behaviour is investigated. For better orientation we now list the key theorems: these are Theorems 1.3.2 and 1.3.5 on hitting times, Theorem 1.4.2 on the strong Markov property, Theorem 1.5.3 characterizing recurrence and transience, Theorem 1.7.7 on invariant distributions and positive recurrence. Theorem 1.8.3 on convergence to equilibrium, Theorem 1.9.3 on reversibility, and Theorem 1.10.2 on long-run averages. Once you understand these you will understand the basic theory. Part of that understanding will come from familiarity with examples, so a large number are worked out in the text. Exercises at the end of each section are an important part of the exposition.

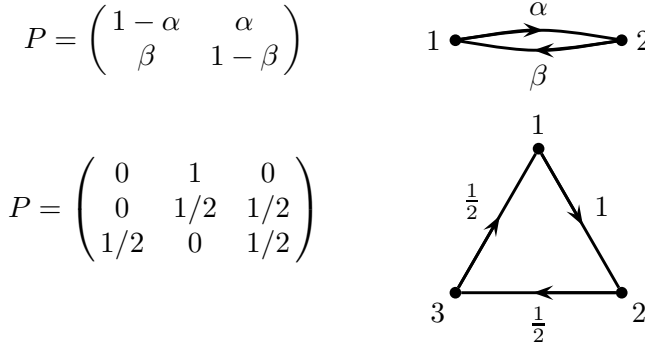
1.1 Definition and basic properties

Let I be a countable set. Each $i \in I$ is called a *state* and I is called the *state-space*. We say that $\lambda = (\lambda_i : i \in I)$ is a *measure* on I if $0 \leq \lambda_i < \infty$ for all $i \in I$. If in addition the *total mass* $\sum_{i \in I} \lambda_i$ equals 1, then we call λ a *distribution*. We work throughout with a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Recall that a *random variable* X with values in I is a function $X : \Omega \rightarrow I$. Suppose we set

$$\lambda_i = \mathbb{P}(X = i) = \mathbb{P}(\{\omega : X(\omega) = i\}).$$

Then λ defines a distribution, the *distribution of X* . We think of X as modelling a random state which takes the value i with probability λ_i . There is a brief review of some basic facts about countable sets and probability spaces in [Chapter 6](#).

We say that a matrix $P = (p_{ij} : i, j \in I)$ is *stochastic* if every row $(p_{ij} : j \in I)$ is a distribution. There is a one-to-one correspondence between stochastic matrices P and the sort of diagrams described in the Introduction. Here are two examples:



We shall now formalize the rules for a Markov chain by a definition in terms of the corresponding matrix P . We say that $(X_n)_{n \geq 0}$ is a *Markov chain* with *initial distribution* λ and *transition matrix* P if

- (i) X_0 has distribution λ ;
- (ii) for $n \geq 0$, conditional on $X_n = i$, X_{n+1} has distribution $(p_{ij} : j \in I)$ and is independent of $X_0, \dots, X_{n-1}$.

More explicitly, these conditions state that, for $n \geq 0$ and $i_1, \dots, i_{n+1} \in I$,

- (i) $\mathbb{P}(X_0 = i_1) = \lambda_{i_1}$;
- (ii) $\mathbb{P}(X_{n+1} = i_{n+1} \mid X_0 = i_1, \dots, X_n = i_n) = p_{i_n i_{n+1}}$.

We say that $(X_n)_{n \geq 0}$ is *Markov*(λ, P) for short. If $(X_n)_{0 \leq n \leq N}$ is a finite sequence of random variables satisfying (i) and (ii) for $n = 0, \dots, N-1$, then we again say $(X_n)_{0 \leq n \leq N}$ is *Markov*(λ, P).

It is in terms of properties (i) and (ii) that most real-world examples are seen to be Markov chains. But mathematically the following result appears to give a more comprehensive description, and it is the key to some later calculations.

Theorem 1.1.1. *A discrete-time random process $(X_n)_{0 \leq n \leq N}$ is Markov(λ, P) if and only if for all $i_1, \dots, i_N \in I$*

$$\mathbb{P}(X_0 = i_1, X_1 = i_2, \dots, X_N = i_N) = \lambda_{i_1} p_{i_1 i_2} p_{i_2 i_3} \dots p_{i_{N-1} i_N}. \quad (1.1)$$

Proof. Suppose $(X_n)_{0 \leq n \leq N}$ is Markov(λ, P), then

$$\begin{aligned} \mathbb{P}(X_0 = i_1, X_1 = i_2, \dots, X_N = i_N) \\ &= \mathbb{P}(X_0 = i_1) \mathbb{P}(X_1 = i_2 \mid X_0 = i_1) \\ &\quad \dots \mathbb{P}(X_N = i_N \mid X_0 = i_1, \dots, X_{N-1} = i_{N-1}) \\ &= \lambda_{i_1} p_{i_1 i_2} \dots p_{i_{N-1} i_N}. \end{aligned}$$

On the other hand, if (1.1) holds for N , then by summing both sides over $i_N \in I$ and using $\sum_{j \in I} p_{ij} = 1$ we see that (1.1) holds for $N - 1$ and, by induction

$$\mathbb{P}(X_0 = i_1, X_1 = i_2, \dots, X_n = i_n) = \lambda_{i_1} p_{i_1 i_2} \dots p_{i_{n-1} i_n}$$

for all $n = 0, 1, \dots, N$. In particular, $\mathbb{P}(X_0 = i_1) = \lambda_{i_1}$ and, for $n = 0, 1, \dots, N - 1$,

$$\begin{aligned} \mathbb{P}(X_{n+1} = i_{n+1} \mid X_0 = i_1, \dots, X_n = i_n) \\ &= \mathbb{P}(X_0 = i_1, \dots, X_n = i_n, X_{n+1} = i_{n+1}) / \mathbb{P}(X_0 = i_1, \dots, X_n = i_n) \\ &= p_{i_n i_{n+1}}. \end{aligned}$$

So $(X_n)_{0 \leq n \leq N}$ is Markov(λ, P). $\square$

The next result reinforces the idea that Markov chains have no memory. We write $\delta_i = (\delta_{ij} : j \in I)$ for the *unit mass* at i , where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 1.1.2 (Markov property). *Let $(X_n)_{n \geq 0}$ be Markov(λ, P). Then, conditional on $X_m = i$, $(X_{m+n})_{n \geq 0}$ is Markov(δ_i, P) and is independent of the random variables $X_0, \dots, X_m$.*

Proof. We have to show that for any event A determined by $X_0, \dots, X_m$ we have

$$\begin{aligned} \mathbb{P}(\{X_m = i_m, \dots, X_{m+n} = i_{m+n}\} \cap A \mid X_m = i) \\ &= \delta_{ii_m} p_{i_m i_{m+1}} \dots p_{i_{m+n-1} i_{m+n}} \mathbb{P}(A \mid X_m = i) \end{aligned} \quad (1.2)$$

then the result follows by Theorem 1.1.1. First consider the case of elementary events

$$A = \{X_0 = i_1, \dots, X_m = i_m\}.$$

In that case we have to show

$$\begin{aligned} & \mathbb{P}(X_0 = i_1, \dots, X_{m+n} = i_{m+n} \text{ and } i = i_m) / \mathbb{P}(X_m = i) \\ &= \delta_{ii_m} p_{i_m i_{m+1}} \cdots p_{i_{m+n-1} i_{m+n}} \\ & \times \mathbb{P}(X_0 = i_1, \dots, X_m = i_m \text{ and } i = i_m) / \mathbb{P}(X_m = i) \end{aligned}$$

which is true by Theorem 1.1.1. In general, any event A determined by $X_0, \dots, X_m$ may be written as a countable disjoint union of elementary events

$$A = \bigcup_{k=1}^{\infty} A_k.$$

Then the desired identity (1.2) for A follows by summing up the corresponding identities for A_k . $\square$

The remainder of this section addresses the following problem: *what is the probability that after n steps our Markov chain is in a given state?* First we shall see how the problem reduces to calculating entries in the n th power of the transition matrix. Then we shall look at some examples where this may be done explicitly.

We regard distributions and measures λ as row vectors whose components are indexed by I , just as P is a matrix whose entries are indexed by $I \times I$. When I is finite we will often label the states $1, 2, \dots, N$; then λ will be an N -vector and P an $N \times N$ -matrix. For these objects, matrix multiplication is a familiar operation. We extend matrix multiplication to the general case in the obvious way, defining a new measure λP and a new matrix P^2 by

$$(\lambda P)_j = \sum_{i \in I} \lambda_i p_{ij}, \quad (P^2)_{ik} = \sum_{j \in I} p_{ij} p_{jk}.$$

We define P^n similarly for any n . We agree that P^0 is the identity matrix I , where $(I)_{ij} = \delta_{ij}$. The context will make it clear when I refers to the state-space and when to the identity matrix. We write $p_{ij}^{(n)} = (P^n)_{ij}$ for the (i, j) entry in P^n .

In the case where $\lambda_i > 0$ we shall write $\mathbb{P}_i(A)$ for the conditional probability $\mathbb{P}(A \mid X_0 = i)$. By the Markov property at time $m = 0$, under $\mathbb{P}_i$, $(X_n)_{n \geq 0}$ is Markov(δ_i, P). So the behaviour of $(X_n)_{n \geq 0}$ under $\mathbb{P}_i$ does not depend on λ .

Theorem 1.1.3. *Let $(X_n)_{n \geq 0}$ be Markov(λ, P). Then, for all $n, m \geq 0$,*

- (i) $\mathbb{P}(X_n = j) = (\lambda P^n)_j$;
- (ii) $\mathbb{P}_i(X_n = j) = \mathbb{P}(X_{n+m} = j \mid X_m = i) = p_{ij}^{(n)}$.

Proof. (i) By Theorem 1.1.1

$$\begin{aligned}\mathbb{P}(X_n = j) &= \sum_{i_1 \in I} \dots \sum_{i_{n-1} \in I} \mathbb{P}(X_0 = i_1, \dots, X_{n-1} = i_{n-1}, X_n = j) \\ &= \sum_{i_1 \in I} \dots \sum_{i_{n-1} \in I} \lambda_{i_1} p_{i_1 i_2} \dots p_{i_{n-1} j} = (\lambda P^n)_j.\end{aligned}$$

(ii) By the Markov property, conditional on $X_m = i$, $(X_{m+n})_{n \geq 0}$ is Markov (δ_i, P) , so we just take $\lambda = \delta_i$ in (i). $\square$

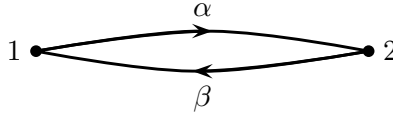
In light of this theorem we call $p_{ij}^{(n)}$ the *n-step transition probability from i to j*. The following examples give some methods for calculating $p_{ij}^{(n)}$.

Example 1.1.4

The most general two-state chain has transition matrix of the form

$$P = \begin{pmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{pmatrix}$$

and is represented by the following diagram:



We exploit the relation $P^{n+1} = P^n P$ to write

$$p_{11}^{(n+1)} = p_{12}^{(n)} \beta + p_{11}^{(n)} (1 - \alpha).$$

We also know that $p_{11}^{(n)} + p_{12}^{(n)} = \mathbb{P}_1(X_n = 1 \text{ or } 2) = 1$, so by eliminating $p_{12}^{(n)}$ we get a recurrence relation for $p_{11}^{(n)}$:

$$p_{11}^{(n+1)} = (1 - \alpha - \beta) p_{11}^{(n)} + \beta, \quad p_{11}^{(0)} = 1.$$

This has a unique solution (see [Section 1.11](#)):

$$p_{11}^{(n)} = \begin{cases} \frac{\beta}{\alpha + \beta} + \frac{\alpha}{\alpha + \beta} (1 - \alpha - \beta)^n & \text{for } \alpha + \beta > 0 \\ 1 & \text{for } \alpha + \beta = 0. \end{cases}$$

Example 1.1.5 (Virus mutation)

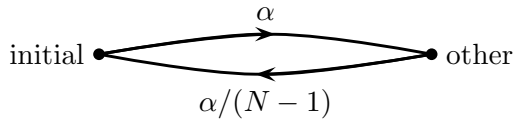
Suppose a virus can exist in N different strains and in each generation either stays the same, or with probability α mutates to another strain, which is chosen at random. What is the probability that the strain in the n th generation is the same as that in the 0th?

We could model this process as an N -state chain, with $N \times N$ transition matrix P given by

$$p_{ii} = 1 - \alpha, \quad p_{ij} = \alpha/(N - 1) \quad \text{for } i \neq j.$$

Then the answer we want would be found by computing $p_{11}^{(n)}$. In fact, in this example there is a much simpler approach, which relies on exploiting the symmetry present in the mutation rules.

At any time a transition is made from the initial state to another with probability α , and a transition from another state to the initial state with probability $\alpha/(N - 1)$. Thus we have a two-state chain with diagram



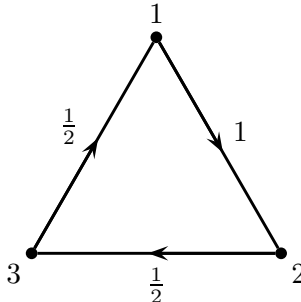
and by putting $\beta = \alpha/(N - 1)$ in Example 1.1.4 we find that the desired probability is

$$\frac{1}{N} + \left(1 - \frac{1}{N}\right) \left(1 - \frac{\alpha N}{N - 1}\right)^n.$$

Beware that in examples having less symmetry, this sort of lumping together of states may not produce a Markov chain.

Example 1.1.6

Consider the three-state chain with diagram



and transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}.$$

The problem is to find a general formula for $p_{11}^{(n)}$.

First we compute the eigenvalues of P by writing down its characteristic equation

$$0 = \det(x - P) = x(x - \frac{1}{2})^2 - \frac{1}{4} = \frac{1}{4}(x - 1)(4x^2 + 1).$$

The eigenvalues are $1, i/2, -i/2$ and from this we deduce that $p_{11}^{(n)}$ has the form

$$p_{11}^{(n)} = a + b \left(\frac{i}{2}\right)^n + c \left(-\frac{i}{2}\right)^n$$

for some constants a, b and c . (The justification comes from linear algebra: having distinct eigenvalues, P is diagonalizable, that is, for some invertible matrix U we have

$$P = U \begin{pmatrix} 1 & 0 & 0 \\ 0 & i/2 & 0 \\ 0 & 0 & -i/2 \end{pmatrix} U^{-1}$$

and hence

$$P^n = U \begin{pmatrix} 1 & 0 & 0 \\ 0 & (i/2)^n & 0 \\ 0 & 0 & (-i/2)^n \end{pmatrix} U^{-1}$$

which forces $p_{11}^{(n)}$ to have the form claimed.) The answer we want is real and

$$\left(\pm \frac{i}{2}\right)^n = \left(\frac{1}{2}\right)^n e^{\pm i n \pi / 2} = \left(\frac{1}{2}\right)^n \left(\cos \frac{n\pi}{2} \pm i \sin \frac{n\pi}{2}\right)$$

so it makes sense to rewrite $p_{11}^{(n)}$ in the form

$$p_{11}^{(n)} = \alpha + \left(\frac{1}{2}\right)^n \left\{ \beta \cos \frac{n\pi}{2} + \gamma \sin \frac{n\pi}{2} \right\}$$

for constants α, β and γ . The first few values of $p_{11}^{(n)}$ are easy to write down, so we get equations to solve for α, β and γ :

$$1 = p_{11}^{(0)} = \alpha + \beta$$

$$0 = p_{11}^{(1)} = \alpha + \frac{1}{2}\gamma$$

$$0 = p_{11}^{(2)} = \alpha - \frac{1}{4}\beta$$

so $\alpha = 1/5$, $\beta = 4/5$, $\gamma = -2/5$ and

$$p_{11}^{(n)} = \frac{1}{5} + \left(\frac{1}{2}\right)^n \left\{ \frac{4}{5} \cos \frac{n\pi}{2} - \frac{2}{5} \sin \frac{n\pi}{2} \right\}.$$

More generally, the following method may in principle be used to find a formula for $p_{ij}^{(n)}$ for any M -state chain and any states i and j .

- (i) Compute the eigenvalues $\lambda_1, \dots, \lambda_M$ of P by solving the characteristic equation.
- (ii) If the eigenvalues are distinct then $p_{ij}^{(n)}$ has the form

$$p_{ij}^{(n)} = a_1 \lambda_1^n + \dots + a_M \lambda_M^n$$

for some constants $a_1, \dots, a_M$ (depending on i and j). If an eigenvalue λ is repeated (once, say) then the general form includes the term $(an + b)\lambda^n$.

- (iii) As roots of a polynomial with real coefficients, complex eigenvalues will come in conjugate pairs and these are best written using sine and cosine, as in the example.

Exercises

1.1.1 Let $B_1, B_2, \dots$ be disjoint events with $\bigcup_{n=1}^{\infty} B_n = \Omega$. Show that if A is another event and $\mathbb{P}(A|B_n) = p$ for all n then $\mathbb{P}(A) = p$.

Deduce that if X and Y are discrete random variables then the following are equivalent:

- (a) X and Y are independent;
- (b) the conditional distribution of X given $Y = y$ is independent of y .

1.1.2 Suppose that $(X_n)_{n \geq 0}$ is Markov (λ, P) . If $Y_n = X_{kn}$, show that $(Y_n)_{n \geq 0}$ is Markov (λ, P^k) .

1.1.3 Let X_0 be a random variable with values in a countable set I . Let $Y_1, Y_2, \dots$ be a sequence of independent random variables, uniformly distributed on $[0, 1]$. Suppose we are given a function

$$G : I \times [0, 1] \rightarrow I$$

and define inductively

$$X_{n+1} = G(X_n, Y_{n+1}).$$

Show that $(X_n)_{n \geq 0}$ is a Markov chain and express its transition matrix P in terms of G . Can all Markov chains be realized in this way? How would you simulate a Markov chain using a computer?